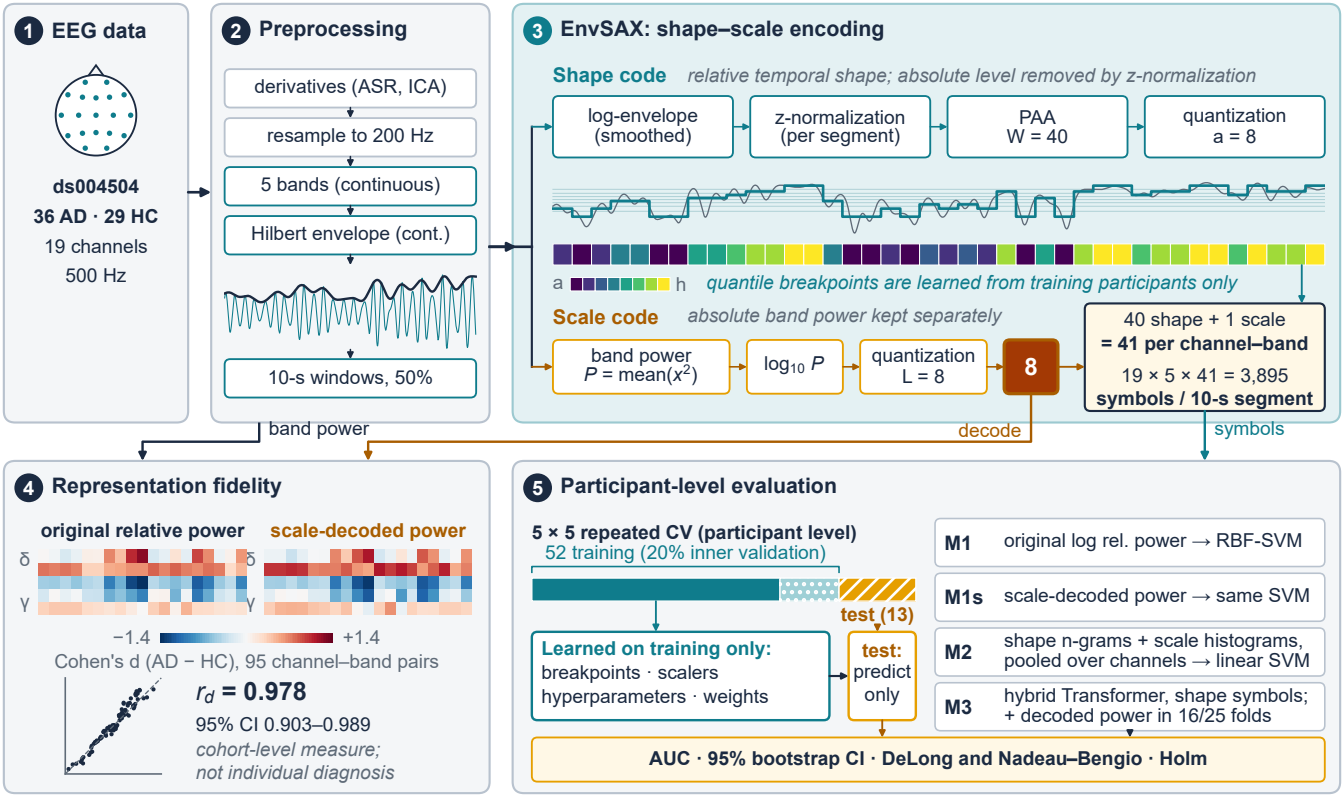

1 EEG data
ds004504
36 AD · 29 HC
19 channels
500 Hz
2 Preprocessing
derivatives (ASR, ICA)
resample to 200 Hz
5 bands (continuous)
Hilbert envelope (cont.)
10-s windows, 50%
band power
3 EnvSAX: shape–scale encoding
Shape code
relative temporal shape; absolute level removed by z-normalization
log-envelope (smoothed)
z-normalization (per segment)
PAA W = 40
quantization a = 8
a
h
quantile breakpoints are learned from training participants only
Scale code
absolute band power kept separately
band power P = mean(x²)
log10 P
quantization L = 8
8
40 shape + 1 scale
= 41 per channel–band
19 × 5 × 41 = 3,895
symbols / 10-s segment
decode
symbols
4 Representation fidelity
original relative power
scale-decoded power
δ
γ
−1.4
+1.4
Cohen's d (AD − HC), 95 channel–band pairs
r_d = 0.978
95% CI 0.903–0.989
cohort-level measure;
not individual diagnosis
5 Participant-level evaluation
5 × 5 repeated CV (participant level)
52 training (20% inner validation)
test (13)
Learned on training only:
breakpoints · scalers
hyperparameters · weights
test:
predict only
M1 original log rel. power → RBF-SVM
M1s scale-decoded power → same SVM
M2 shape n-grams + scale histograms, pooled over channels → linear SVM
M3 hybrid Transformer, shape symbols; + decoded power in 16/25 folds
AUC · 95% bootstrap CI · DeLong and Nadeau–Bengio · Holm